

Firing of coal and biomass and their mixtures in 50 kW and 12 MW circulating fluidized beds “Phenomenon study and comparison of scales

The [original paper](#) contains 10 sections, with 9 passages identified by our machine learning algorithms as central to this paper.

Paper Summary

SUMMARY PASSAGE 1

Introduction

The combustion of coal in circulating fluidized-bed (CFB) boilers has been reviewed [1]. There is a vast amount of literature on coal and, more recently, also on biomass combustion in CFB boilers. Some papers also deal with co-combustion of coal and biomass or waste in steady-state operation [2][3][4][5][6].

SUMMARY PASSAGE 2

Experimental

The reactivity of the fuels and fuel mixtures was studied by analysing responses to step changes in the fuel feed. Fuel ratio changes were also carried out for the coal and wood mixtures. The tests discussed above were performed using both test reactors and are examined here, see Fig. 1.

SUMMARY PASSAGE 3

Discussion

In order to understand the boiler dynamics and recognise factors which affect the process dynamics, it is important to carry out and analyse diverse process experiments. The dynamic behaviour when using different fuels and fuel mixtures and their combustion in two different sized reactors is compared. Study of the steady-state combustion reveals odd features concerning the co-combustion process.

SUMMARY PASSAGE 4

Steady-State Analyses

The burning regimes model [31] has been presented earlier and was used to analyse combustion regimes of coal, wood chips and mixtures of these fuels. The operation of the model is based on the idea that a burning fuel particle initially loses its mass due to drying and devolatilization, i.e. its average density decreases, but its diameter remains approximately constant. The analysis was carried out for different size fractions and summed up to find the locations of the release of volatiles and char combustion.

SUMMARY PASSAGE 5

Burning In Bed Region

The moisture content of wood is substantially higher than that of coals. Coal with higher density contains a great amount of char and combustion occurs mainly in the bed area. Wood with a lower density does not stay in the bed but releases moisture there on route to the freeboard.

SUMMARY PASSAGE 6

Burning In Circulation

As discussed, wood chips release moisture in the bed area, yet wood devolatilization also occurs, which consumes oxygen from the bed. The measured oxygen profiles and the result from the burning regimes model, shown in Figs. 8 and 3, support this conclusion. Low temperature and oxygen concentration in the bed slow the combustion of the coal char considerably which increases char inventory in the bed.

SUMMARY PASSAGE 7

Experiments On Combustion Dynamics

However, when comparing the responses of the two reactors, it can be seen that in this case dynamics of the outlet responses are close to each other when using fuels containing large amounts of volatiles i.e. wood chips and coal-wood mixtures. The temperature in the bed area and upper part of the reactor during step change in the fuel feed are shown in Fig. 12 combustion between the two reactors. However, the changes in temperatures were similar for both reactors also for coal.

SUMMARY PASSAGE 8

Time-Dependent Models And Analyses

Combustion dynamics can be studied in different scales, where fuel reactivity and the process dynamics significantly differ. The combustion rate of the fuel is not the only important matter to be considered, but also the way in which the heat is released. For example, biofuels contain Chalm.

SUMMARY PASSAGE 9

Conclusions

Measurements with coal, wood chips and mixture of coal and wood chips showed that temperature profiles with burning of mixtures cannot be deduced directly from the burning profiles of individual fuel components. The emphasis of the co-firing trials was on the dynamic changes. Also the dynamic responses for mixtures can be surprisingly different from the responses when burning of individual fuels.